

Harpreet Singh

AI Intern

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EDUCATION

•Bachelor of Technology in Computer Science Engineering

Lovely Professional University, Phagwara, Punjab

Sept 2022 – Aug 2026

CGPA: 8.15 / 10.0

EXPERIENCE

•AI Backend Intern

FlyRank AI

Jul 2026 – Present

Remote

- Built **Python** backend services integrating **LLMs with REST APIs, databases, and external applications**, powering production AI workflows.
- Implemented **multi-step LLM workflows and tool-calling** patterns for AI-powered automation, collaborating cross-functionally to improve response accuracy and system reliability.

PROJECTS

•Multi-Agent Research Assistant – Autonomous Agentic Workflow

Python, LangChain, LLMs, Tool Calling, Web Scraping

May 2026 – Jul 2026

GitHub

- Developed an **autonomous multi-agent system** in **Python** using **LangChain** and **LLMs** to automate research workflows — web search, content extraction, report generation, and review.
- Orchestrated a **4-agent architecture** (search, read, write, critique) with **tool calling**, improving task decomposition and enabling modular agentic workflow design.
- Integrated **LLMs with external APIs** (Tavily), BeautifulSoup, and Streamlit into an end-to-end application with real-time retrieval, structured output, and an interactive UI.

•EduVerse AI — Agentic University RAG Assistant

Python, FastAPI, LangChain, ChromaDB, Ollama, Groq

Jan 2026 – Mar 2026

GitHub

- Architected a **modular agentic RAG system** with **hierarchical sub-agent orchestration** in LangChain, integrating LLMs with a **vector database** (ChromaDB / pgvector); achieved **91% retrieval accuracy** with zero external API dependencies.
- Implemented **multi-turn conversational context management** with LangChain memory; built a multi-format ETL pipeline improving retrieval precision by **over 30%** vs. keyword baseline.
- Applied **anti-hallucination prompting** with cosine-similarity thresholding, cutting out-of-context outputs to under **7%**; optimized the **FastAPI** pipeline to **sub-800ms average latency**.

•Language Identification System — Multilingual NLP Classifier

TensorFlow, CNN, LSTM, Naive Bayes, Deep Learning

May 2025 – Jun 2025

GitHub

- Designed a multilingual text classification system supporting **17 languages** using CNN, LSTM, and Naive Bayes with systematic **hyperparameter tuning** and cross-validation; used confusion-matrix error analysis to improve robustness on noisy text.
- Achieved **96.52% classification accuracy**; research **published** at the 2025 Grenze International Conference and indexed in the **Grenze International Journal of Engineering and Technology (Scopus)**.
- Benchmarked against baseline **TF-IDF and n-gram classifiers**, demonstrating a **12% accuracy gain**; optimized preprocessing (tokenization, normalization) for noisy, code-mixed, short-text inputs at scale.

TECHNICAL SKILLS

Languages	Python, Java, SQL
AI Agents & MCP	AI Agents, Agentic Workflows, Agent Design, Tool Calling, MCP (Model Context Protocol), Multi-step Automation
ML & AI	LLMs, RAG, NLP, Deep Learning, Prompt Engineering, A/B Prompt Testing, Hyperparameter Tuning
LLM Platforms	OpenAI GPT-3.5/4, Google Gemini API, Anthropic Claude API, n8n, Ollama
Frameworks	LangChain, LangGraph, FastAPI, PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, Sentence-Transformers, spaCy, NumPy, Pandas, XGBoost
Tools	Firebase, Docker, Git, GitHub, REST APIs, Streamlit
Databases	PostgreSQL (pgvector), ChromaDB
Core CS	Data Structures & Algorithms, Operating Systems, DBMS, OOP

ACHIEVEMENTS & AWARDS

- **Marvell M-STEM Scholar (Top 100 Winner, 2026)**: Selected among the Top 100 engineering students nationwide for the Marvell Scholarship for Technical and Engineering Merit (M-STEM) under the Excellence category, recognizing exceptional academic performance, technical excellence, and leadership potential.
- **Research Publication (Scopus-Indexed)**: Peer-reviewed paper on multilingual NLP classification published in the Grenze International Journal of Engineering and Technology (2025), indexed in **Scopus**.
- **AI Training Expert, Outlier (Remote)**: Evaluated LLM code-generation quality across **GPT and Claude model families** to support alignment and instruction-following improvements.
- **Google Nibbler AI Safety Project**: Selected among **120 students from 8,000+ applicants** to evaluate adversarial image-generation systems and identify safety vulnerabilities in production-grade AI models.

CERTIFICATIONS

- **Complete Agentic AI Bootcamp with LangGraph & LangChain** — Udemy Aug 2026
- **Java Software Development** — Coursera Nov 2025
- **Machine Learning Specialization** — Andrew Ng, Coursera Jul 2024
- **Generative AI with Large Language Models** — Coursera Mar 2024